



Understanding Cloud Computing

Introduction

This document provides a comprehensive overview of cloud computing, its key concepts, benefits, service models, and popular providers. Refer to the video at [00:00:00] for more information.

What is Cloud Computing?

Cloud computing is the delivery of computing services—including servers, storage, databases, networking, software, analytics, and intelligence—over the Internet (“the cloud”) to offer faster innovation, flexible resources, and economies of scale. Instead of using your local computer or a private data center, you access these resources via the internet. Refer to the video at [00:00:12] for more information.

Example: Google Drive and Google Photos

Services like Google Drive and Google Photos exemplify cloud storage. You can store your photos and files on these platforms and access them from anywhere without needing to save them on your local devices. This is because the storage is available over the internet, not on your personal machine. Refer to the video at [00:00:24] for more information.

Key Concepts

1. **On-Demand Scalability:** The ability to increase or decrease the size of your cloud resources based on demand. For example, you can scale from 2 GB to 2 TB of data storage as needed. Refer to the video at [00:00:46] for more information.
2. **Scalability:** The capacity to scale up or down based on the number of users or the volume of traffic. If you suddenly need to support 100,000 users instead of 100, the cloud allows you to scale your system accordingly. Refer to the video at [00:00:54] for more information.
3. **Pay-as-you-go:** You only pay for the resources you use. There are no upfront costs for hardware, and you don't pay for resources you're not actively using. This model is similar to paying a utility bill. Refer to the video at [00:01:06] for more information.



Why Companies Use the Cloud

Cost Saving

- **No Upfront Hardware Cost:** Companies don't need to invest in expensive hardware infrastructure. Refer to the video at [00:01:25] for more information.
- **Pay-as-you-go Model:** You only pay for the cloud services you use, such as EC2 or S3 on AWS. Refer to the video at [00:01:28] for more information.
- **Lower Maintenance Cost:** Cloud providers handle hardware maintenance and updates, reducing the burden on the company. Refer to the video at [00:01:36] for more information.

Scalability

- **Instant Scalability:** Cloud resources can be scaled up or down instantly based on demand. This is particularly useful for handling traffic spikes, such as during popular events on streaming services like Jio Hotstar. Refer to the video at [00:01:41] for more information.
- **Handling Traffic Spikes:** The cloud can handle a large number of users without downtime, making it ideal for businesses experiencing rapid growth or traffic fluctuations. Refer to the video at [00:01:53] for more information.

Faster Time to Market

- **Quick Deployment:** Applications can be deployed quickly using cloud services like AWS Lambda or Google Firebase. Refer to the video at [00:01:57] for more information.
- **Pre-built Tools:** Cloud providers offer pre-built tools that speed up development, allowing companies to bring their products to market faster. Refer to the video at [00:02:03] for more information.

Top Cloud Providers

There are several top cloud providers to choose from, including: Refer to the video at [00:02:11] for more information.

- Alibaba Cloud
- Amazon Web Services (AWS)
- Microsoft Azure
- Dell
- IBM
- Dropbox
- Salesforce



Cloud Service Models

Cloud computing offers several service models, each providing a different level of control and responsibility. Refer to the video at [00:02:22] for more information.

Software as a Service (SaaS)

SaaS delivers software applications over the internet. The provider fully manages the application, and users simply access it. Examples include Microsoft 365 and Google Workspace (Gmail, Google Sheets). Users don't need to worry about the underlying infrastructure; they just use the service. Refer to the video at [00:02:27] for more information.

Platform as a Service (PaaS)

PaaS provides a platform for developers to build, run, and manage applications without managing the underlying infrastructure. Heroku, Azure App Service, and Google App Engine are examples of PaaS. Developers can focus on coding and deploying applications without worrying about server maintenance. Refer to the video at [00:02:48] for more information.

Infrastructure as a Service (IaaS)

IaaS provides basic computing infrastructure, such as servers, storage, and networks, on demand. Users have more control over the infrastructure but are responsible for managing it. Amazon EC2 is a popular IaaS offering, allowing users to create and manage their own virtual servers. Refer to the video at [00:03:12] for more information.

Popular Cloud Providers in Detail

Amazon Web Services (AWS)

AWS offers over 200 services and is a market leader in cloud computing. Its strengths include a massive ecosystem, global presence, and scalability. AWS is a good starting point for those new to cloud computing. Refer to the video at [00:03:32] for more information.

Microsoft Azure

Developed by Microsoft, Azure is strong in hybrid cloud and enterprise integrations. It is enterprise-friendly and well-suited for .NET developers. Services include Azure Virtual Machines, App Services, and Cosmos DB. Refer to the video at [00:03:44] for more information.



Summary of Cloud Service Models

- **SaaS:** Everything is managed by the provider; you just use the service (e.g., Google Photos). Refer to the video at [00:04:04] for more information.
- **PaaS:** You use the provider's platform to build and run your applications. Refer to the video at [00:04:12] for more information.
- **IaaS:** You use the provider's infrastructure to create and manage your own servers and networks. Refer to the video at [00:04:16] for more information.

IaaS offers the most control but requires more management, while SaaS offers the least control but requires minimal management. PaaS falls in between, providing a balance of control and ease of use. Refer to the video at [00:04:22] for more information.

Reflection Questions

1. Can you describe a scenario where using cloud computing would be more beneficial than using a traditional on-premises data center? Explain your reasoning.
2. How do the different cloud service models (SaaS, PaaS, IaaS) cater to different needs and levels of technical expertise? Provide examples of when you might choose one over the others.

Answer Key

1. **Scenario:** A startup company that needs to quickly launch a new application with minimal upfront investment. **Reasoning:** Cloud computing eliminates the need for purchasing and maintaining expensive hardware, allowing the startup to focus on development and deployment while scaling resources as needed.
2. **SaaS:** Best for end-users who need ready-to-use applications without managing infrastructure (e.g., using Google Docs for word processing). **PaaS:** Ideal for developers who want to build and deploy applications without managing servers or networking (e.g., using Heroku to deploy a web application). **IaaS:** Suitable for organizations that need full control over their infrastructure and want to manage servers, storage, and networking (e.g., using AWS EC2 to create custom virtual machines).